

ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION TECHNOLOGY DEPARTMENT

UNIT: Manage Computerized Database Systems

UNIT CODE: 0612 451 08A

LEVEL: 5

PRACTICAL ASSESSMENT

TIME: 3 HOURS

TOTAL MARKS: 50 MARKS

PRACTICAL TASK: STUDENT RECORD MANAGEMENT DATABASE

A college requires a computerized system for storing and managing student records. You are required to create a Microsoft Access database that will allow the college to store student information, enter records through a form and produce reports.

Materials and Equipment

No.	Material/Equipment	Quantity
1	Computer desktop	1
2	Microsoft Access	1
3	Keyboard and mouse	1 set
4	Power supply	1

Instructions to Candidates

1. Read and understand all the instructions before starting the practical.
2. Use the **computer desktop** provided to you to complete the practical.
3. Use **Microsoft Access** to create and manage the database.
4. Create a new database and save it as **StudentRecords_YourName.accdb**.
5. Store your work in the location specified by the trainer/assessor.
6. Create the table structure exactly as specified.
7. Use the **form** to enter the student records. Do not enter the records directly into the table after creating the form.
8. Create the required reports using the data stored in the database.
9. Save each database object using the specified name.
10. Check your database before submission to ensure that all records, forms and reports are available.

11. Do not delete or modify the database of another candidate.
12. Ask the assessor for clarification if you do not understand an instruction.
13. Submit the completed **.accdb** file at the end of the practical.

TASK A: CREATE THE DATABASE AND TABLE STRUCTURE

1. Start **Microsoft Access**.
2. Create a blank database.
3. Save the database as:

StudentRecords_YourName.accdb

4. Create a table named **Students** using the following structure:

Field Name	Data Type	Description
StudentID	AutoNumber	Primary key
AdmissionNo	Short Text	Student admission number
FullName	Short Text	Student full name
Gender	Short Text	Male/Female
Course	Short Text	Course undertaken
PhoneNumber	Short Text	Student telephone number
DateOfBirth	Date/Time	Student date of birth
YearOfStudy	Number	Current year of study

5. Set **StudentID** as the **Primary Key**.
6. Save the table as **Students**.

TASK B: CREATE A FORM AND ENTER RECORDS

Create a form using the **Students** table.

Save the form as:

frmStudentRegistration

Use the form to enter the following records:

AdmissionNo	FullName	Gender	Course	PhoneNumber	DateOfBirth	YearOfStudy
ATC001	Brian Mwangi	Male	ICT	0712345678	15/03/2005	1
ATC002	Mary Wanjiku	Female	ICT	0723456789	21/07/2004	2
ATC003	John Kamau	Male	Computer Science	0734567890	10/01/2005	1
ATC004	Jane Njeri	Female	ICT	0745678901	18/11/2003	3
ATC005	Peter Kariuki	Male	Computer Science	0756789012	25/05/2004	2
ATC006	Lucy Achieng	Female	ICT	0767890123	30/09/2005	1
ATC007	David Ochieng	Male	Information Technology	0778901234	12/02/2003	3
ATC008	Ann Wambui	Female	Information Technology	0789012345	08/06/2004	2
ATC009	Samuel Maina	Male	ICT	0790123456	14/12/2005	1
ATC010	Grace Muthoni	Female	Computer Science	0701234567	22/08/2003	3

The form should:

- Display the student fields clearly.
- Have a suitable title.
- Allow entry of student records.
- Allow movement between records.
- Have a neat and user-friendly layout.

TASK C: GENERATE STUDENT RECORDS REPORT

Create a report based on the **Students** table.

The report should display:

- Admission Number
- Full Name
- Gender
- Course
- Phone Number
- Year of Study

Give the report the title:

STUDENT RECORDS REPORT

Save it as:

rptStudentRecords

Format the report neatly so that the information can be easily read.

TASK D: GENERATE STUDENTS BY COURSE REPORT

Create a second report that groups students according to their **Course**.

The report should:

1. Group students by course.
2. Display the Admission Number.
3. Display the Full Name.
4. Display the Course.
5. Use a suitable report title.
6. Have a clear and professional layout.

Save the report as:

rptStudentsByCourse

TASK E: VERIFY AND SAVE THE DATABASE

Before submitting your work:

1. Open the **Students** table and confirm that all 10 records are available.
2. Open **frmStudentRegistration** and confirm that it displays the student records correctly.
3. Open **rptStudentRecords** and confirm that the report displays the required information.
4. Open **rptStudentsByCourse** and confirm that students are correctly grouped by course.
5. Save and close the database.
6. Confirm that the database file is saved as:

StudentRecords_YourName.accdb

Required Database Objects

Object	Required Name
Table	Students
Form	frmStudentRegistration
Report	rptStudentRecords
Report	rptStudentsByCourse